

Diagnosing Conformal Prediction Failures Under Distribution Shift

A COVID-19 Case Study

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KEY FINDING: SHAP Concentration predicts CP failure in GBMs ($\rho = 0.853$)

40% threshold achieves 87.5% LOO-CV accuracy | GBM models only

SHAP Concentration Method

Definition:

$$\text{SHAP Conc.} = \frac{\varphi_1}{\sum_{j=1}^p |\varphi_j|}$$

φ_1 = Top feature SHAP; $\sum |\varphi_j|$ = Total SHAP

Decision Threshold (40%):

< 40% Robust
≥ 40% Vulnerable

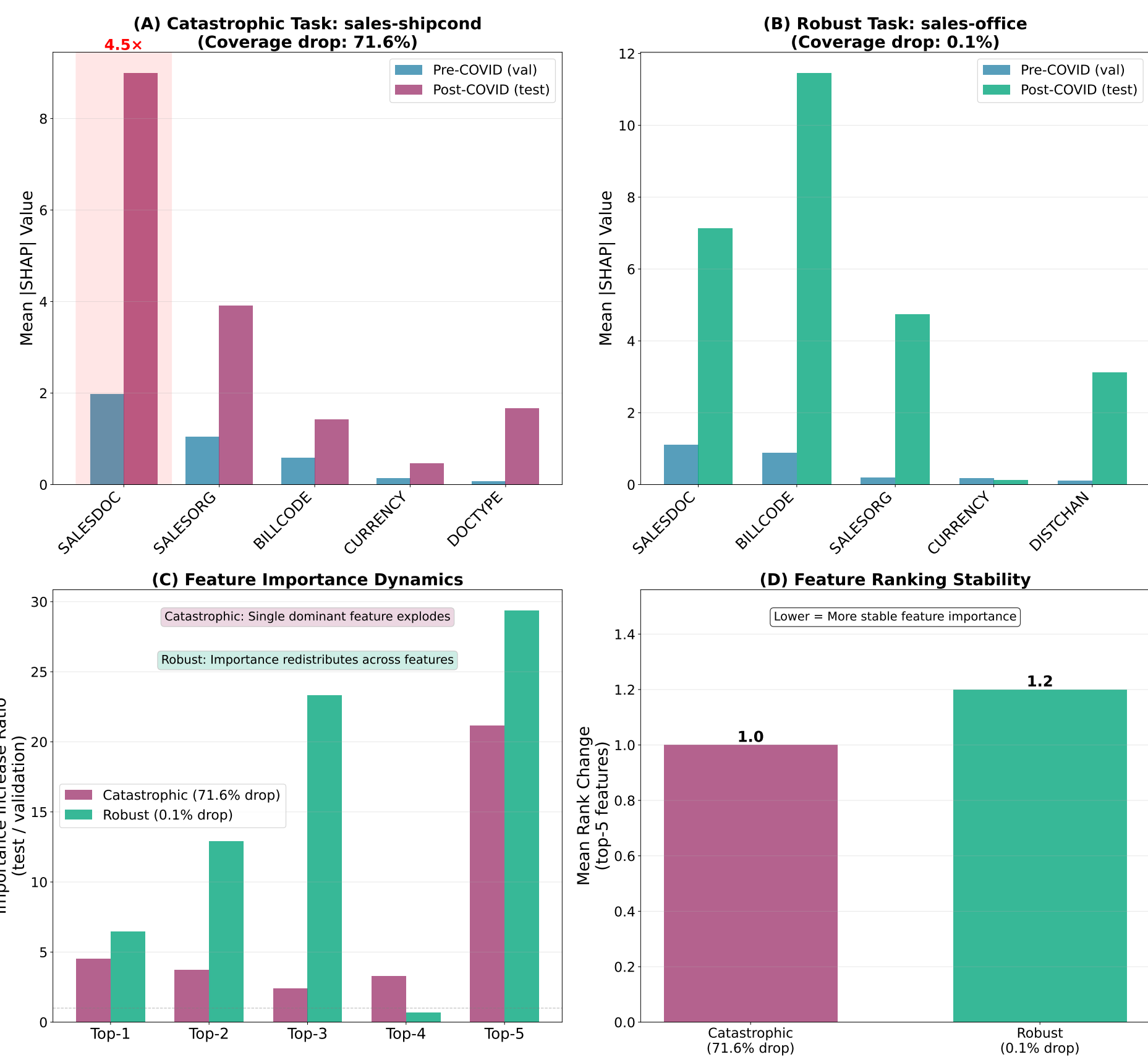


Fig 2: Feature importance concentration

Problem & Motivation

Conformal Prediction (CP) provides distribution-free uncertainty, but **distribution shift breaks guarantees**.

Research Question:

Can we **predict** when CP fails **before** deployment?

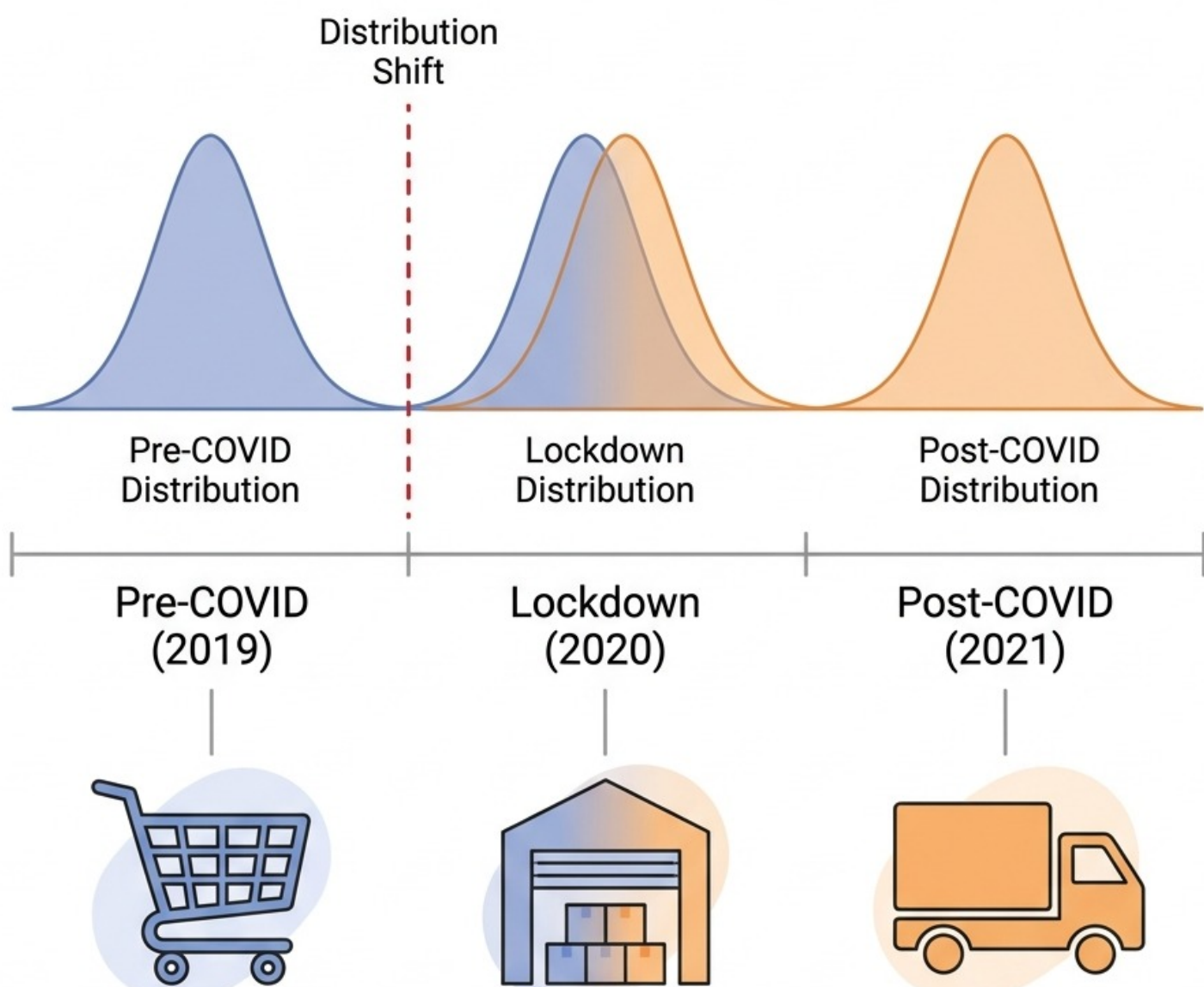


Fig 1: COVID-19 distribution shifts (Feb-Jul 2020)

Key Insight

Models with **concentrated feature reliance** are vulnerable:

- Single feature drift $\Rightarrow$ large impact
- No redundant features for backup
- Calibration misalignment

High Concentration
 $\Downarrow$
High Vulnerability

SALT Supply Chain Dataset

RelBench Supply Chain Benchmark

- 8 classification tasks
- COVID-19 temporal split
- Feb-Jul 2020 (onset $\rightarrow$ peak)

Task Type	Count
Sales-level	5
Item-level	3
Total SALT	8
+ External	+8

MAIN RESULTS

Strong Correlation: SHAP Concentration vs Coverage Drop

$$\rho = 0.853$$

Spearman Correlation

$p < 0.001$ | $n = 16$ tasks | 9 domains

SALT COVID (n=8): $\rho = 0.833$ ($p = 0.010$)

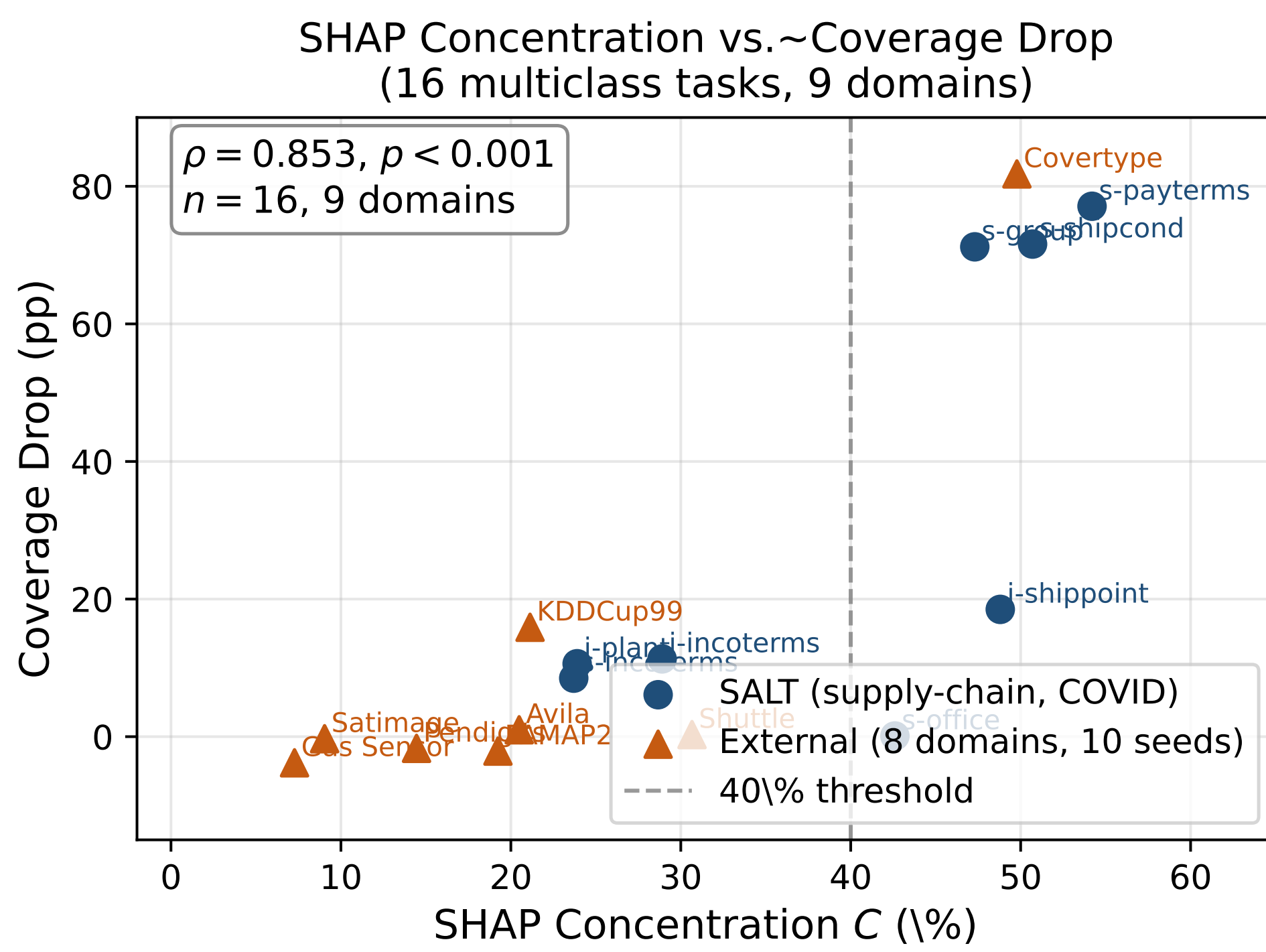


Fig 3: SHAP Concentration vs Coverage Drop (n=16)

Model Scope: GBM Only

Model	ρ	Works?
LightGBM	0.833	✓
CatBoost	0.667	✓
XGBoost	0.548	✓
Random Forest	0.30	✗
MLP	0.43	✗

GBM: concentrated dependence
RF/MLP: global-sensitivity (different failure mode)

External Validation: WILDS + UCI

4 Benchmark Datasets

Dataset	Conc.	Cov.Δ	Pred.
CivilComments	26.2%	-0.7pp	Robust
Amazon	1.9%	+0.4pp	Robust
Covertypes	12.5%	+8.0pp	Robust
Adult	27.5%	+1.4pp	Robust

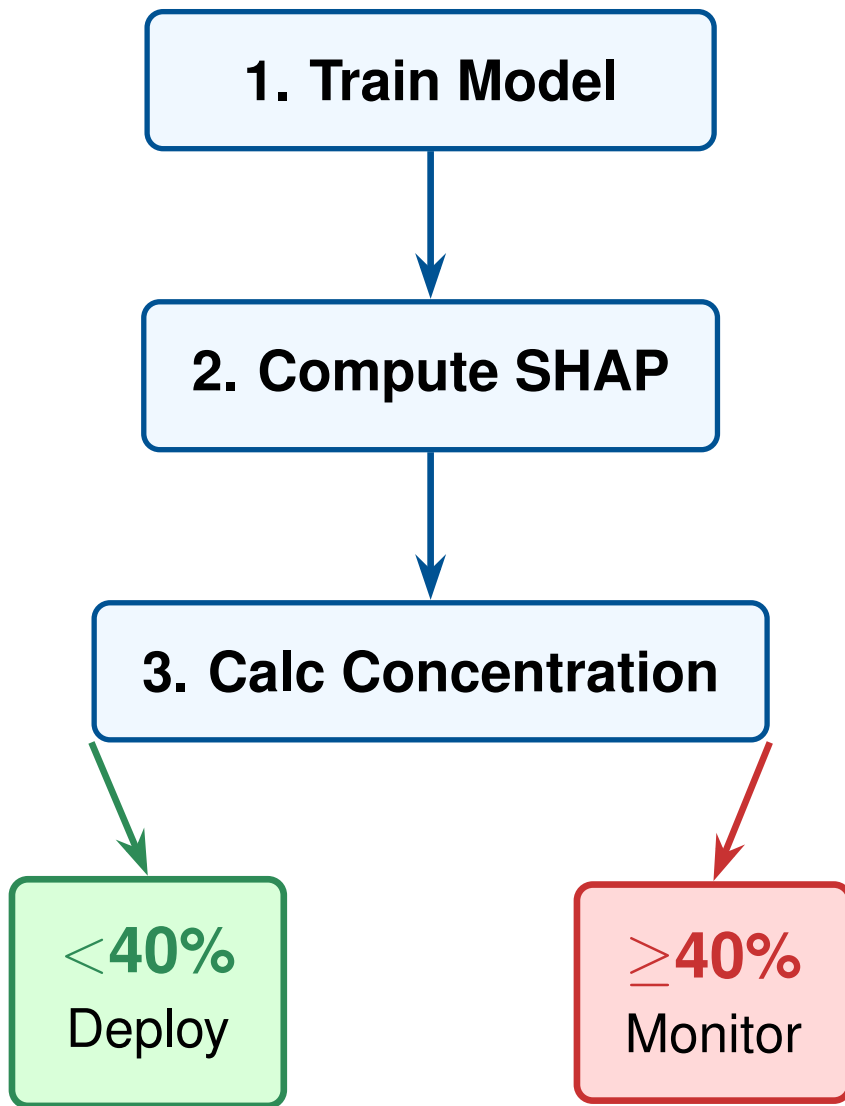
Conc.=SHAP Concentration, Cov.Δ=Coverage Change, Pred.=Prediction

91.7%

Combined Accuracy (11/12)

Practical Framework

Pre-Deployment Assessment



Conclusions

Contributions:

1. SHAP Concentration predicts CP failures in GBMs
2. Strong correlation: $\rho = 0.853$ ($p < 0.001$)
3. 40% threshold: LOO-CV 87.5%, Cohen's d=3.08
4. WILDS validation: 91.7% combined

Scope: GBM only (not RF/MLP)

Future: Adaptive CP, real-time monitoring

Paper & Code:

github.com/chorok-lee/conformal-covid

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